

COURSE: CHE 312: Transport Phenomena II

Fall Semester 2025

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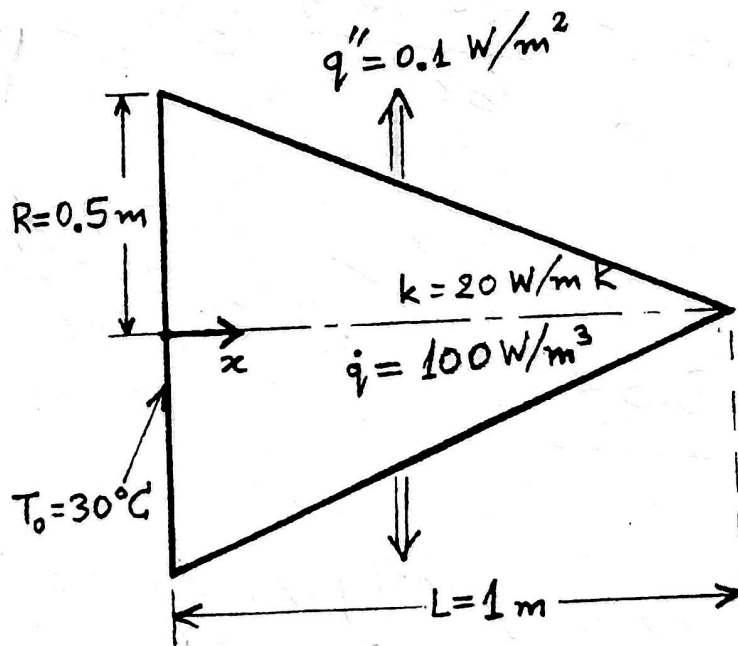
Quiz #2

September 30, 2025

Open Notes and Book

Duration: 40 minutes

A solid cone as shown is of circular cross section and the radius of its base is 0.5 m. The cone of length 1 m and thermal conductivity 20 W/m·K has uniform volumetric heat generation of 100 W/m³. A constant heat flux of 0.1 W/m² is leaving the cone through the lateral surface of the cone, while the end surface at $x = 0$ is maintained at $T_0 = 30^\circ\text{C}$. Determine the temperature profile $T(x)$ and the temperature $T(L)$.



$$L = 1\text{m} \quad T_0 = 30^\circ\text{C}, 303\text{K} \quad k = 20 \frac{\text{W}}{\text{m}\cdot\text{K}}$$

$$q'' = 0.1 \text{ W/m}^2 \quad R = 0.5 \text{ m} \quad x=0 \quad x=L$$

For Flat
surfaces

$$T(x)_{x=0} = T_0 + \frac{q(L^2 - x^2)}{6k}$$

$$= 303\text{K} + \frac{100((1)^2 \text{ m}^2 - (0)^2)}{6(20) \frac{\text{W}}{\text{m}\cdot\text{K}}}$$

$$= 303\text{K} + \frac{5}{6} \text{ K}$$

$$\boxed{= 303.83\text{K}}$$

$$T(x)_{x=L} = T_0 + \frac{q(L^2 - L^2)}{6k} = T_0 + 0$$

$$= T_0$$

$$\boxed{T(x)_{x=L} = 303\text{K}}$$

$$T_0 + \frac{q(0)}{6k} = T_0 + 0$$

$$\frac{d}{dx} \left(k A(x) \frac{dT}{dx} \right) + q''' A(x) - q'' \frac{ds}{dx} = 0 \quad \text{For cone} *$$

$$A(x) = \pi R_0^2 \left(1 - \frac{x}{L}\right)^2$$

$$\frac{ds}{dx} = 2\pi R_0 \left(1 - \frac{x}{L}\right) g \quad g = \sqrt{1 + R_0/L^2}$$

$$r(x) = R_0 \left(1 - \frac{x}{L}\right)$$

$$A(x) = \pi r(x)^2 = \pi R_0^2 \left(1 - \frac{x}{L}\right)^2$$

$$P(x) = 2\pi r(x) = 2\pi R_0 \left(1 - \frac{x}{L}\right)$$

$$g = \sqrt{1 + (R_0/L)^2} = \sqrt{1 + 0.5^2} = \sqrt{1.25} = 1.118$$

$$T(x) = 303 + \frac{q'' L g}{k R_0} - \frac{q'''}{6k} \left(L x - \frac{x^2}{2} \right)$$

$$\downarrow$$

$$T(x) = 303 + \frac{q'' L g}{k R_0} - \frac{q'''}{3k} \left(L x - \frac{x^2}{2} \right)$$

$$T(x) = 303 + \frac{(0.1)(1)^2 \cdot 1.118}{20 \cdot 0.5} - \frac{100}{120}$$

$x=L$

$$= \boxed{302.18 \text{ K}}$$